App Overview

PennyPals is a financial literacy application designed for students in grades 4–6. The app focuses on developing foundational financial skills, including identifying needs vs. wants, basic budgeting, saving, and responsible spending.

The application aligns learning outcomes with interactive activities by ensuring that each lesson directly reinforces a specific concept. For example, the drag-and-drop activity reinforces classification skills (needs vs wants), while the budget builder activity develops decision-making and financial planning skills.

Rather than relying on passive reading, PennyPals uses active learning strategies such as interaction, feedback, and gamification to improve understanding and retention.

The user interface is designed to be simple, colorful, and engaging for younger users, minimizing unnecessary steps and making navigation intuitive.

Installation Guide

Option 1: Using the weblink to access the website (add link after hosting)

Option 2: Download and host locally

1. Open this github repo link https://github.com/seyonr/education-app-project.git (Image 1)
2. Download as zip (Image 2)
3. Open vs code
4. Open downloaded folder in vs code under file tab (Image 3)
5. Open vs code terminal under terminal tab (Image 4)
6. Type `node -v` and press Enter
    a. If version is provided go to next step (Image 5)
    b. If a version is not displayed, go to https://nodejs.org/en/download and download version compatible for your computer specs
7. Type `npm install` and press enter (Image 6)
8. Type `npm start` and click enter (Image 7)

Use Cases

1) Use Case Name: Completing assessment for unit 1

This use case demonstrates how the app reinforces learning through assessment and feedback, allowing students to identify mistakes and improve understanding.

Actor: Grade 4 student named Bill
Goal: Successful completion of assessment

Preconditions:

Unit 1 lesson already done

Main Flow:

1. Bill clicks assessment button on home page (Image 7)
2. Bill selects unit 1 assessment (Image 8)
3. Bill goes through questions, and completes the assessment
4. Bill looks over his mistakes (Image 9)
5. Bill is credited with 20 coins for successful completion

Alternative Flows:
Unit 1 Assessment locked → Bill completes all lessons for unit 1 before trying assessment

Postconditions:

- 20 coins credited
- Progress bar filled

2) Use Case Name: Completing a Lesson
This use case highlights how interactive activities allow students to actively engage with financial concepts rather than passively consuming information.

Actor: Grade 5 student named Sarah
Goal: Learn a financial concept

Main Flow:

- Sarah navigates to the Lessons page
- Selects a lesson
- Completes interactive activities
- Receives feedback and earns 10 coins

Postconditions:
● Lesson marked complete
● 10 Coins awarded

Universal Design for Learning (UDL) Support

- **Multiple Means of Representation**: Concepts are presented through interactive activities such as drag-and-drop and budgeting simulations instead of relying on long text explanations. Visual elements such as icons and simple layouts support younger learners.

- **Multiple Means of Engagement:** Gamification features such as coins, rewards, and progress bars motivate students to continue learning. For example, completing lessons and assessments rewards coins, encouraging consistent participation.

- **Multiple Means of Action and Expression:** Students demonstrate understanding through different activity types, including multiple choice questions, drag-and-drop classification, and budgeting tasks. This allows students with different learning preferences to succeed.

Current Features:

- **Lessons Page**: The Lessons Page provides structured learning modules where each lesson focuses on a specific financial concept. Lessons include multiple types of interactive activities such as multiple choice questions, drag-and-drop classification, and budgeting simulations. These activities promote active learning by requiring students to apply concepts rather than passively read content. Immediate feedback is provided after each interaction, allowing students to correct mistakes and reinforce understanding.

- **Quiz Page**: After completing all lessons in a unit, students complete a quiz to assess their understanding. The quiz reinforces learning by testing key concepts and providing feedback on incorrect answers. Rewards such as coins and progress tracking further motivate students to engage with assessments.

The current prototype has been tested for stability, and no major bugs were observed during normal usage.

Future Features:

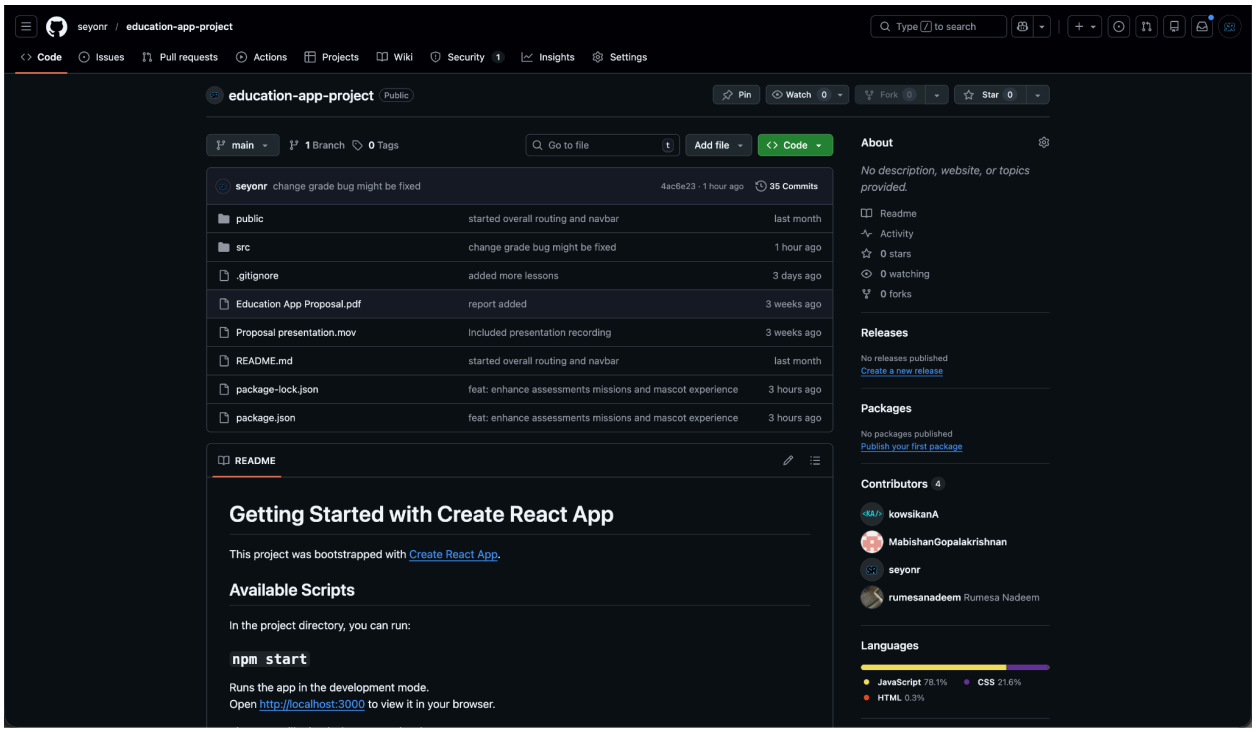
- **Pet Store system:** A gamification feature that allows students to use their earned coins to purchase items for their virtual pet.

- **Animations and Visual Enhancements:** Additional animations will be added throughout the app to make the experience more interactive and appealing for young users

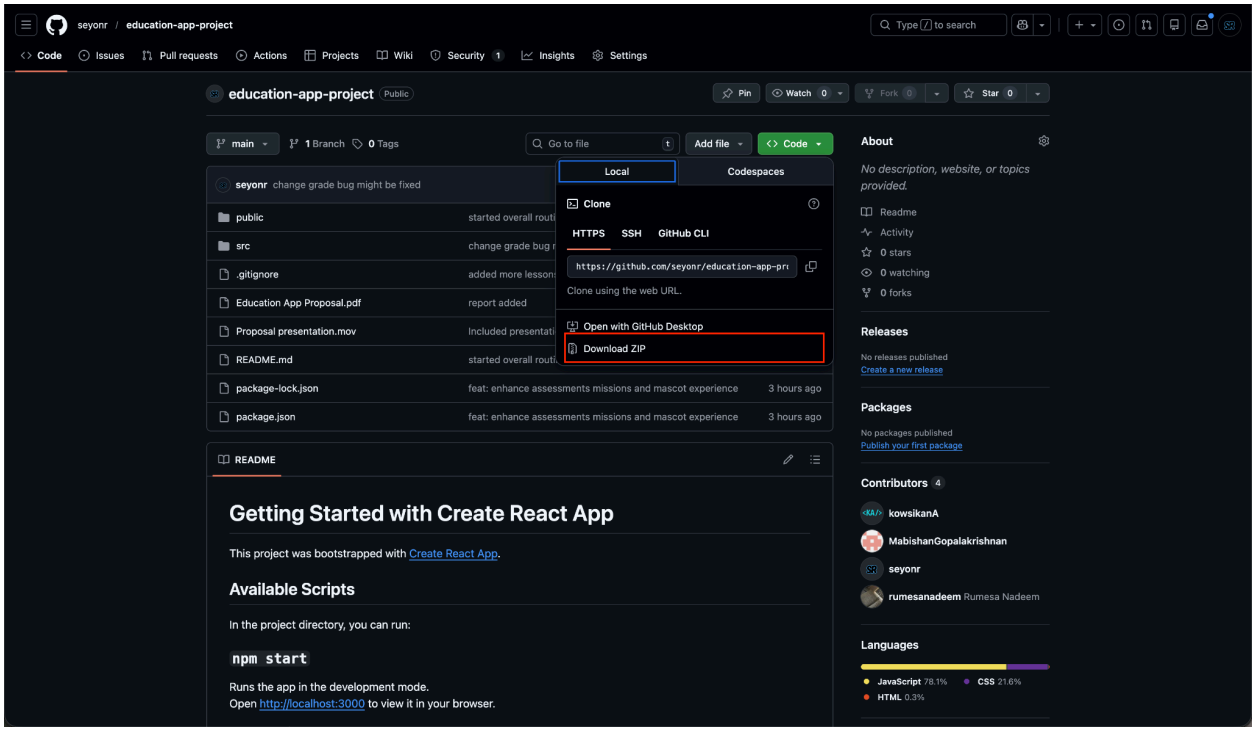
- **Investment Game:** An investment game will be added where users can simulate purchasing stocks and introduce basic concepts of saving and investing

- **Enhance lesson engagement:** Additional improvements will be made to increase lesson engagement

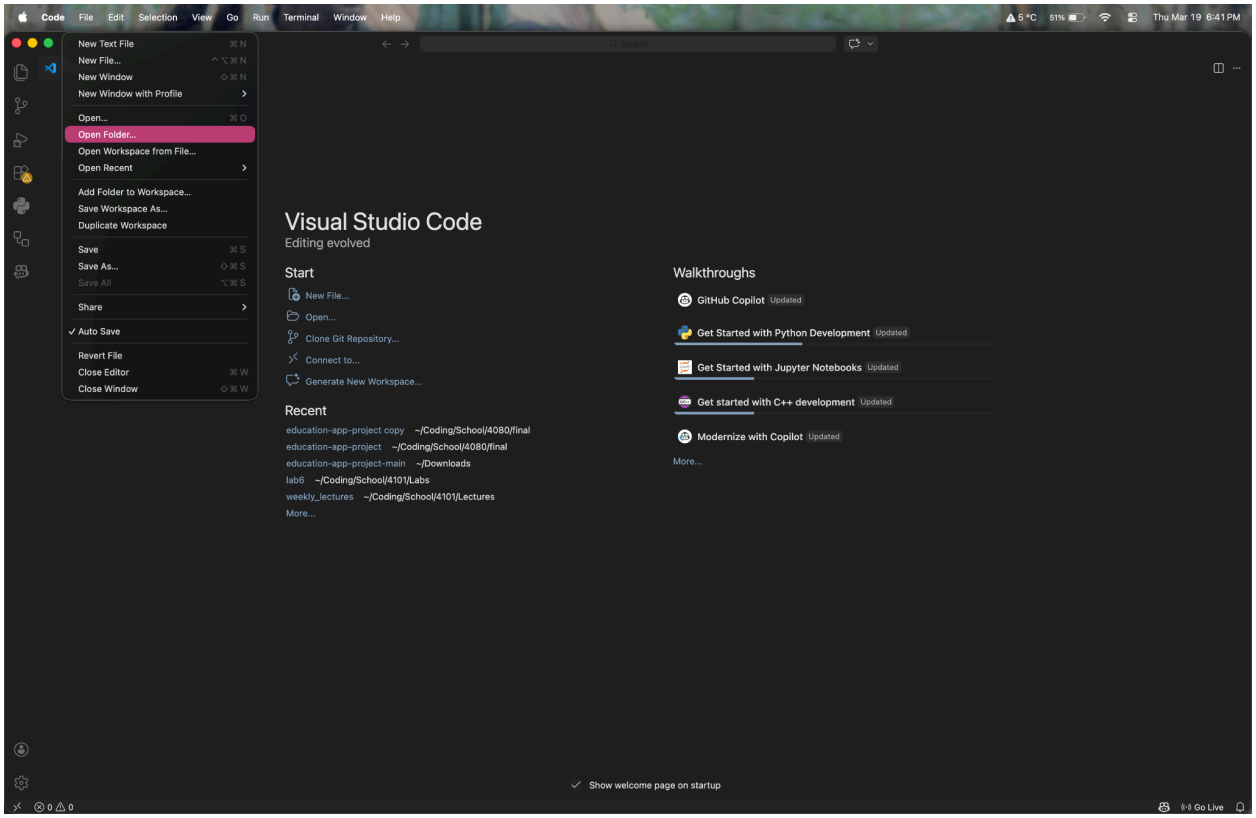
Images



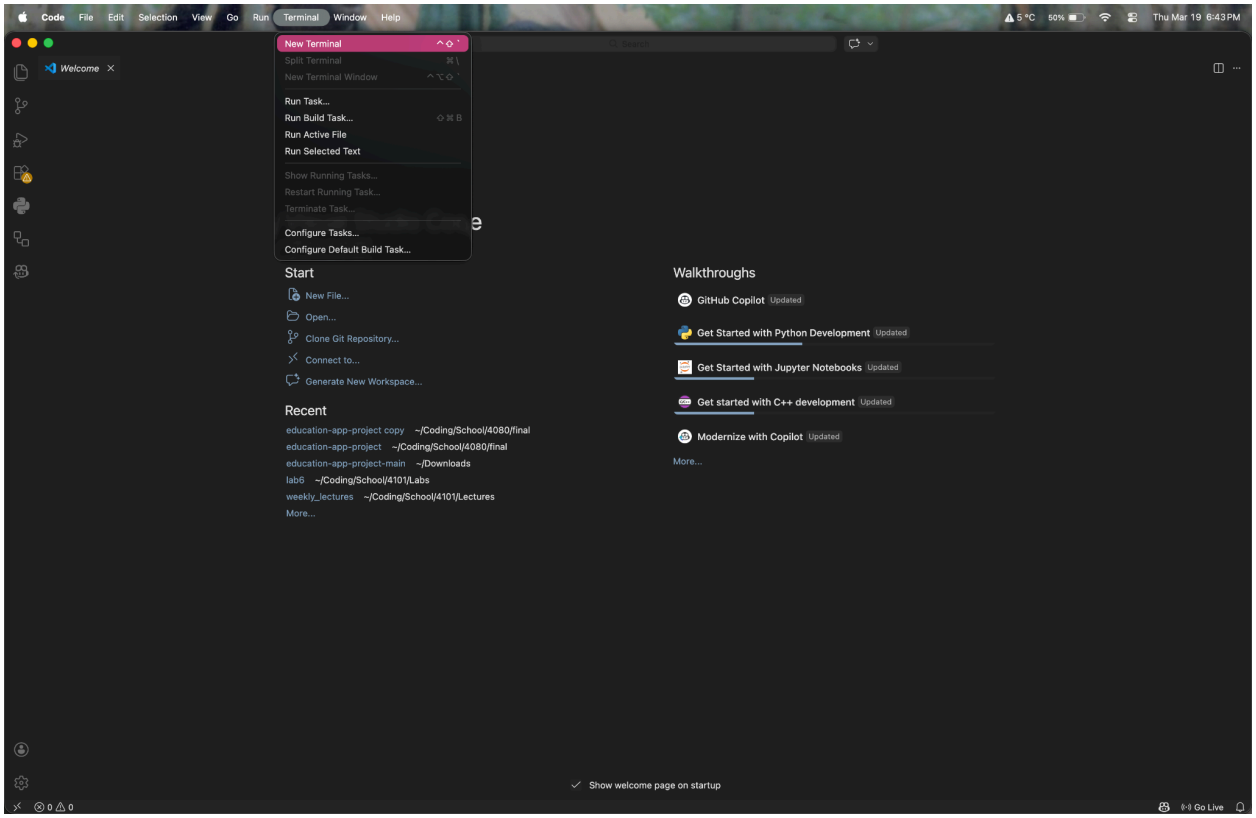
(Image 1)



(Image 2)


Code
File
Edit
Selection
View
Go
Run
Terminal
Window
Help
Thu Mar 19 6:41 PM
New Text File
New File...
New Window
New Window with Profile
Open...
Open Folder...
Open Workspace from File...
Open Recent
Add Folder to Workspace...
Save Workspace As...
Duplicate Workspace
Save
Save As...
Save All
Share
Auto Save
Revert File
Close Editor
Close Window
Visual Studio Code
Editing evolved
Start
New File...
Open...
Clone Git Repository...
Connect to...
Generate New Workspace...
Recent
education-app-project copy ~/Coding/School/4080/final
education-app-project ~/Coding/School/4080/final
education-app-project-main ~/Downloads
lab6 ~/Coding/School/4101/Labs
weekly_lectures ~/Coding/School/4101/Lectures
More...
Walkthroughs
GitHub Copilot Updated
Get Started with Python Development Updated
Get Started with Jupyter Notebooks Updated
Get started with C++ development Updated
Modernize with Copilot Updated
More...
Show welcome page on startup
Go Live

(Image 3)


Code File Edit Selection View Go Run Terminal Window Help
Thu Mar 19 6:43 PM
Welcome
New Terminal
Split Terminal
New Terminal Window
Run Task...
Run Build Task...
Run Active File
Run Selected Text
Show Running Tasks...
Restart Running Task...
Terminate Task...
Configure Tasks...
Configure Default Build Task...
Start
New File...
Open...
Clone Git Repository...
Connect to...
Generate New Workspace...
Recent
education-app-project copy ~/Coding/School/4080/final
education-app-project ~/Coding/School/4080/final
education-app-project-main ~/Downloads
lab6 ~/Coding/School/4101/Labs
weekly_lectures ~/Coding/School/4101/Lectures
More...
Walkthroughs
GitHub Copilot Updated
Get Started with Python Development Updated
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Get started with C++ development Updated
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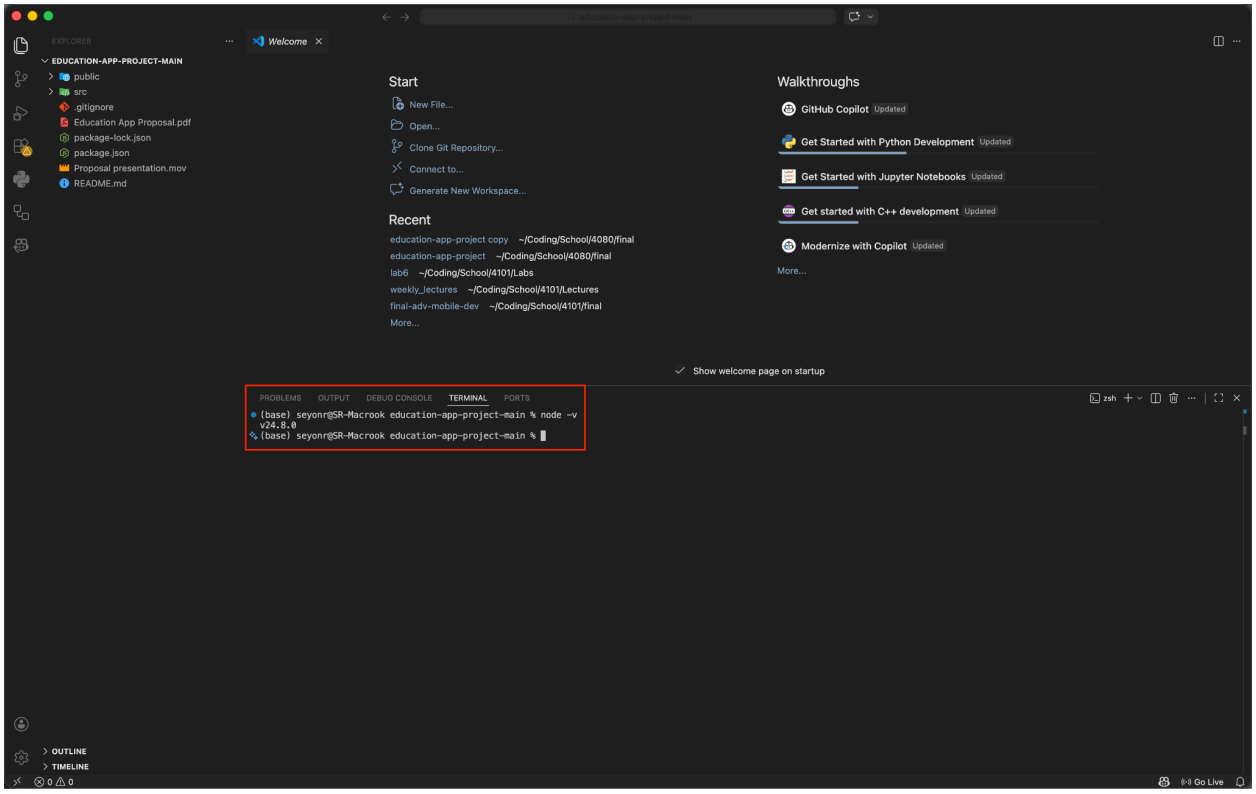
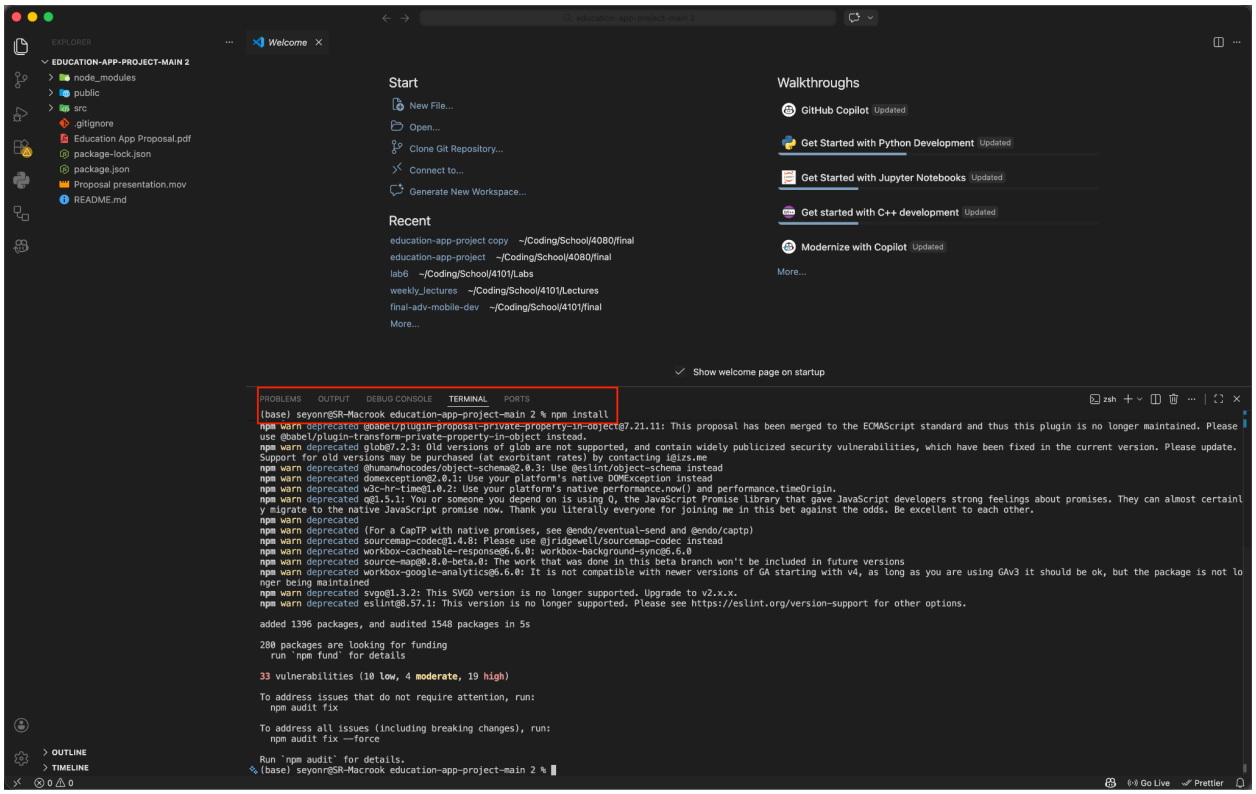
More...
Show welcome page on startup
Go Live

(Image 4)


EXPLORER
EDUCATION-APP-PROJECT-MAIN
public
src
.gitignore
Education App Proposal.pdf
package-lock.json
package.json
Proposal presentation.mov
README.md
Welcome
Start
New File...
Open...
Clone Git Repository...
Connect to...
Generate New Workspace...
Recent
education-app-project copy ~/Coding/School/4080/final
education-app-project ~/Coding/School/4080/final
lab6 ~/Coding/School/4101/Labs
weekly_lectures ~/Coding/School/4101/Lectures
final-adv-mobile-dev ~/Coding/School/4101/final
More...
Walkthroughs
GitHub Copilot Updated
Get Started with Python Development Updated
Get Started with Jupyter Notebooks Updated
Get started with C++ development Updated
Modernize with Copilot Updated
More...
Show welcome page on startup
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) seyonr@SR-Macrook education-app-project-main % node -v
v24.8.0
(base) seyonr@SR-Macrook education-app-project-main %
OUTLINE
TIMELINE
Go Live

(Image 5)



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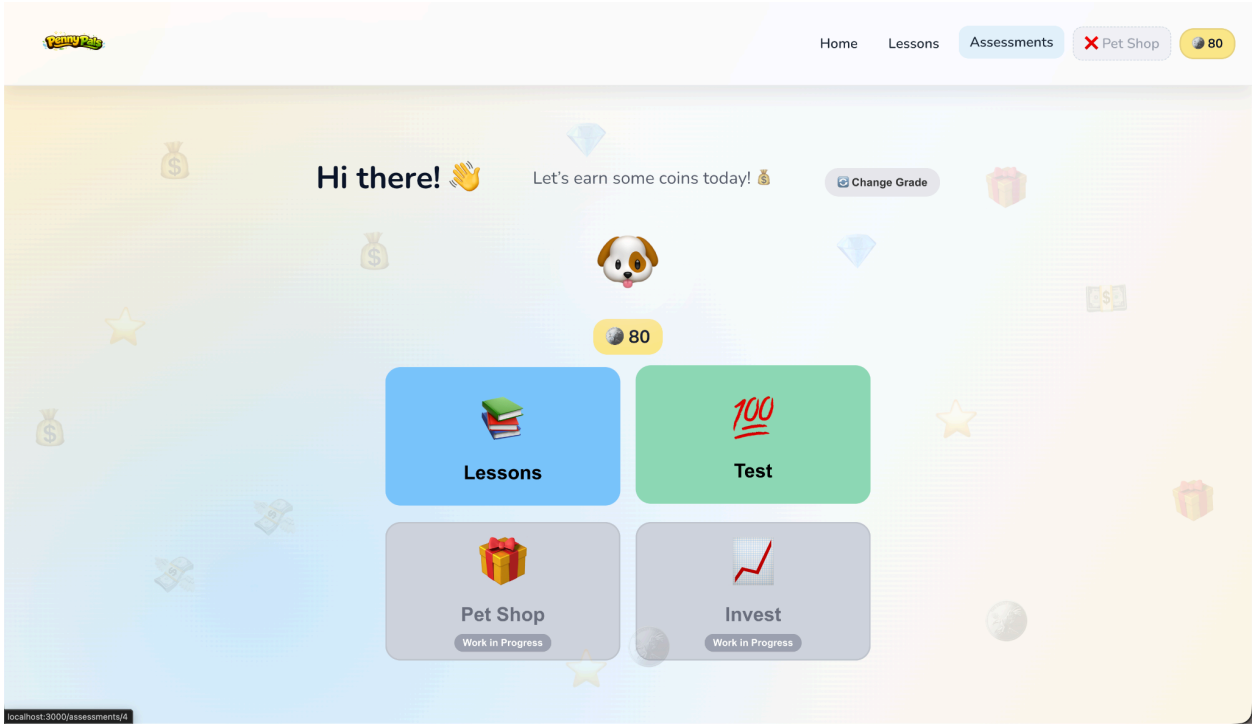
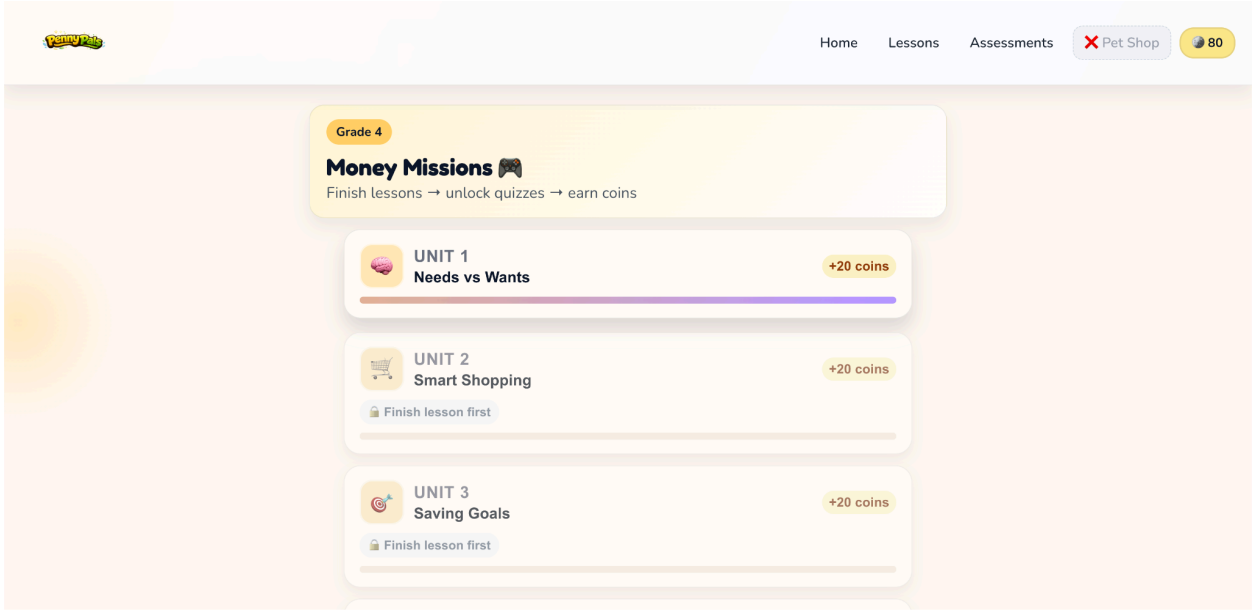
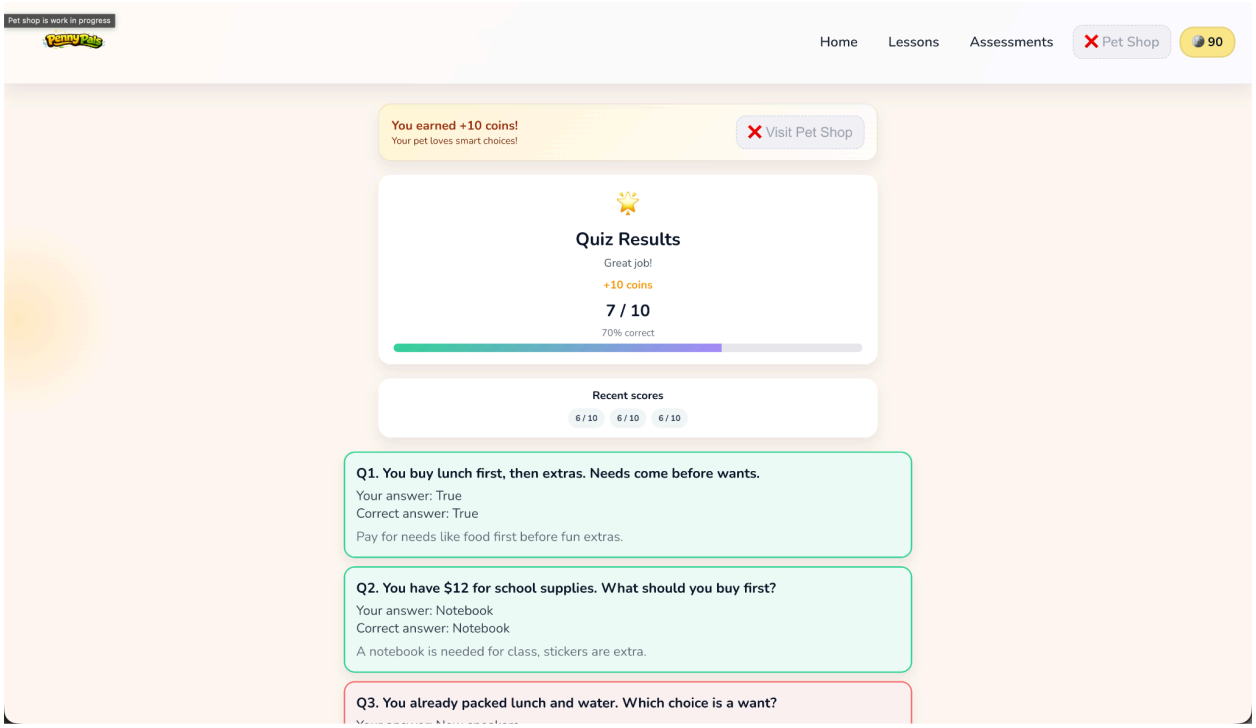


(image 7)



(Image 8)


Pet shop is work in progress
Home
Lessons
Assessments
Pet Shop
90
You earned +10 coins!
Your pet loves smart choices!
Visit Pet Shop
Quiz Results
Great job!
+10 coins
7 / 10
70% correct
Recent scores
6 / 10
6 / 10
6 / 10
Q1. You buy lunch first, then extras. Needs come before wants.
Your answer: True
Correct answer: True
Pay for needs like food first before fun extras.
Q2. You have $12 for school supplies. What should you buy first?
Your answer: Notebook
Correct answer: Notebook
A notebook is needed for class, stickers are extra.
Q3. You already packed lunch and water. Which choice is a want?

(Image 9)